

Response of functionally graded cylindrical shells under moving thermo-mechanical loads

P. Malekzadeh*, Y. Heydarpour

Department of Mechanical Engineering, Persian Gulf University, Bushehr 75168, Iran

ARTICLE INFO

Article history:

Received 27 October 2011

Accepted 16 April 2012

Available online 22 May 2012

Keywords:

Functionally graded materials

Cylindrical shells

Thermoelasticity

Moving mechanical and thermal loads

Finite element method

Differential quadrature method

ABSTRACT

The transient thermoelastic analysis of functionally graded (FG) cylindrical shells under moving boundary pressure and heat flux is presented. The material properties are assumed to be temperature-dependent and graded in the radial direction. The hyperbolic heat conduction equations are used to include the influence of finite heat wave speed (i.e., the non-Fourier effect). To benefit from the high accuracy and low computational efforts of the differential quadrature method (DQM) in conjunction with the effectiveness of the finite element method (FEM) in general geometry, loading and systematic boundary treatment, a combination of these methods is employed to discretize the governing equations in the spatial domain. The resulting system of differential equations is solved using Newmark's time integration scheme in the temporal domain. The presented formulation and method of solution are validated by showing their fast rate of convergence and by comparing the results, in the limit cases, with those obtained using the commercial finite element package ANSYS and some other available solutions in the literature. Then, the effects of different geometrical, material and load parameters on the transient thermoelastic behavior of the FG cylinders under moving mechanical and thermal loads are studied.

© 2012 Elsevier Ltd. All rights reserved.

1. Introduction

Functionally graded materials (FGMs) are a relatively new class of materials whose compositions are designed to change continuously within the body, particularly along the thickness direction. Usually, these materials are made from a mixture of ceramic and metal. Structural members made of FGMs have found wide applications in modern industries such as mechanical, aerospace, nuclear engineering and reactors. Hence, their dynamic characteristics under thermo-mechanical loading are of great interest for engineering design and manufacture.

The static, buckling, free vibration and dynamic analysis of FG cylindrical shells under stationary mechanical and thermal loading have been investigated by many researchers; see for example Refs. [1–10]. In comparison to the research works in these topics, only limited studies can be found on the analysis of FG cylindrical shells subjected to moving heat source or mechanical loads [11–16].

Ootao et al. [11] studied the transient temperature and thermal stress distribution in an infinitely long nonhomogeneous hollow cylinder due to a moving heat source in the axial direction from the inner and/or outer surfaces using the layerwise theory in conjunction with the method of Fourier cosine and Laplace

transformations. Hasheminejad and Komeili [12] presented axisymmetric response analysis of an arbitrarily thick, isotropic, and infinitely long FG circular cylindrical shell subjected to an axially moving normal ring load. The problem solution is derived by using Fourier transformation with respect to a moving reference frame in conjunction with the transfer matrix solution technique. Jabbari et al. [13] developed an analytical solution for the one-dimensional temperature distribution, mechanical and thermal stresses in an infinitely long FG hollow cylinder under a moving heat source, which moves across the thickness of the cylinder. Based on the first-order shear deformation theory (FSDT), Sheng and Wang [14] prepared an approximate solution for FG laminated piezoelectric cylindrical shells under moving mechanical loads. The modal analysis technique and Newmark's time integration method were used to calculate the dynamic response of FG laminated cylindrical shells. In another work, they studied the effect of thermal shock and moving mechanical loads on the response of FG laminated piezoelectric cylindrical shells [15]. Sofiyev [16] presented an analytical solution for the dynamic behavior of the infinitely-long FG cylindrical shell subjected to combined action of the axial tension, internal compressive load and moving ring-shaped compressive pressure with constant velocity. The problem was studied on the basis of the classical cylindrical shell theory.

To the best of authors' knowledge, there is no result for the thermoelastic behavior of short FG cylindrical shells under moving thermo-mechanical and loads. Therefore, this subject is

* Corresponding author. Tel.: +98 771 4222150; fax: +98 771 4540376.

E-mail addresses: p_malekz@yahoo.com,
malekzadeh@pgu.ac.ir (P. Malekzadeh).

studied here by employing a hybrid numerical method composed of the finite element method (FEM), differential quadrature method (DQM) and finite difference method. The DQM has simple formulation, low computational cost and high accuracy when it is used for the global behavior analysis of structural elements such as buckling and vibrations [9,10,20–23]. Also, it has been successfully employed for different structural problems with variable coefficients' differential equations [9,10,20–23]. Hence, using the DQM along the graded direction enables one to accurately and efficiently discretize the governing differential equations of the FG cylindrical shells and implement exactly the related boundary conditions in this direction. In addition, since in using this method the strong form of the governing equations and the related boundary conditions are discretized in the thickness direction (the graded direction), it is free of the shear locking phenomenon that occurred when working with the conventional FEM for beams, plates and shells. On the other hand, the effectiveness of the FEM in general geometry, loading and systematic boundary treatment, especially for moving loads problems have been well known [17–19]. Therefore, by using a combination of DQM and FEM, the advantages of both methods will be achieved. The problem formulations are based on the hyperbolic heat transfer equations for evaluating the temperature distribution and elasticity theory for determining the displacement and stress components in FG cylinders. After demonstrating the correctness of the formulation together with the convergence and accuracy of the method of solution, the effects of different parameters on the response of FG cylinders subjected to moving boundary thermo-mechanical loadings are studied.

2. Governing equations

Consider an FG hollow cylinder which on its inner surface is subjected to a pressure and distributed heat flux with moving upper boundaries as shown in Fig. 1. Its outer surface is free from mechanical loading but is cooled by convection heat transfer. Both pressure and heat flux is assumed to have axisymmetric distribution. Due to axisymmetric geometry, material properties, thermal loading and boundary conditions, the thermoelastic equations reduce to the two-dimensional ones. Consequently, a cylindrical coordinate system (r, z) is sufficient to label the

material points of the cylinder in the undeformed, uniform temperature reference configuration. The governing equations include the hyperbolic heat transfer equation and elasticity equations for evaluating the temperature distribution, displacement and stress tensor components of the FG cylindrical shells, respectively.

The material properties are assumed to vary continuously through the radial direction according to power law distribution in terms of the volume fraction of the constituents. The material composition continuously varies such that the inner surface of the cylinder is ceramic rich whereas the outer surface of the cylinder is metal rich. Based on the power law distribution, a typical effective material property ' P ' of the FG hollow cylinder is obtained as [5],

$$P(r, T) = P_c(T) + [P_m(T) - P_c(T)] \left(\frac{r - R_i}{R_o - R_i} \right)^p \quad (1)$$

where the subscripts m and c refer to the metal and ceramic constituents, respectively; p is the power law index (or the material property graded index), which is a positive real number; R_i and R_o are the inner and outer radius of the hollow cylinder, respectively; T is the temperature at an arbitrary material point of the cylinder. Clearly, when $p=0$, one obtains $P=P_m$ and when p is very large P inclined to P_c .

The temperature dependence of a typical material property ' Q ' of the FG shell can be represented as [5],

$$Q(T) = Q_0(Q_{-1}T^{-1} + 1 + Q_1T + Q_2T^2 + Q_3T^3) \quad (2)$$

The coefficients Q_i ($i = -1, 0, 1, 2, 3$) are unique to the constituent materials.

The constitutive relations for a linear elastic isotropic FG cylindrical shell can be written as

$$\begin{Bmatrix} \sigma_{rr} \\ \sigma_{\theta\theta} \\ \sigma_{zz} \\ \sigma_{rz} \end{Bmatrix} = \begin{bmatrix} C_{11} & C_{12} & C_{12} & 0 \\ C_{12} & C_{11} & C_{12} & 0 \\ C_{12} & C_{12} & C_{11} & 0 \\ 0 & 0 & 0 & C_{55} \end{bmatrix} \begin{Bmatrix} \varepsilon_{rr} - \varepsilon_T \\ \varepsilon_{\theta\theta} - \varepsilon_T \\ \varepsilon_{zz} - \varepsilon_T \\ 2\varepsilon_{rz} \end{Bmatrix} \quad (3)$$

where σ_{ij} and ε_{ij} ($ij=r, \theta, z$) are the components of stress and strain tensor, respectively; $\varepsilon_T (= \int_{T_0}^T \alpha dT)$ is the thermal strain; $\alpha [= \alpha(r, T)]$ is the thermal expansion coefficient; T_0 is the initial temperature and it is assumed that the shell is stress free at this

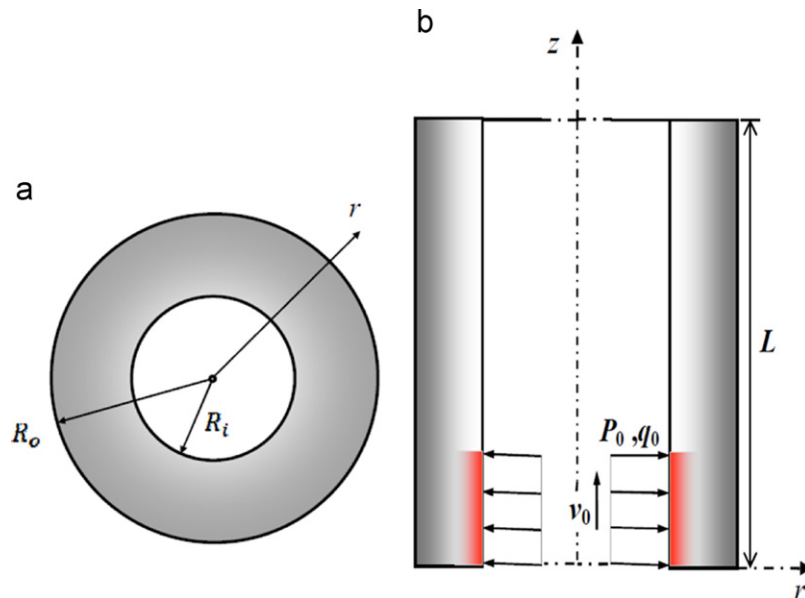


Fig. 1. Geometry and thermal loading of FG hollow cylinder.

temperature; $C_{ij} = C_{ij}(r, T)$; $i, j = 1, 2, 5$ are the material elastic stiffness, which are related to the elastic material properties as follows:

$$C_{11} = \frac{(1-\nu)E}{(1+\nu)(1-2\nu)}, \quad C_{12} = \frac{\nu E}{(1+\nu)(1-2\nu)}, \quad C_{55} = \frac{E}{2(1+\nu)}, \quad (4a-c)$$

where E and ν are Young's modulus and Poisson's ratio, respectively.

Based on the linear elasticity theory, the axisymmetric strain-displacement relations in cylindrical coordinate system become,

$$\varepsilon_{rr} = \frac{\partial u}{\partial r}, \quad \varepsilon_{\theta\theta} = \frac{u}{r}, \quad \varepsilon_{zz} = \frac{\partial w}{\partial z}, \quad \varepsilon_{rz} = \frac{1}{2} \left(\frac{\partial u}{\partial z} + \frac{\partial w}{\partial r} \right) \quad (5a-d)$$

where u and w are the displacement components of an arbitrary material point of the FG cylinder in the r and z -directions, respectively.

The hyperbolic heat transfer equation and the equations of motion can be summarized as, respectively,

Table 1

Temperature-dependent coefficients of material properties for ceramic (Zirconia) and metals (Stainless steel) [5].

	Material	Q_{-1}	Q_0	Q_1	Q_2	Q_3
E	Stainless steel	0	201.04 (GPa)	3.079×10^{-4}	-6.534×10^{-7}	0
	Zirconia	0	244.27 (GPa)	-1.371×10^{-3}	1.214×10^{-6}	-3.681×10^{-10}
ν	Stainless steel	0	0.3262	-2.00×10^{-4}	3.797×10^{-7}	0
	Zirconia	0	0.2882	1.133×10^{-4}	0	0
ρ	Stainless steel	0	8166 (kg/m ³)	0	0	0
	Zirconia	0	5700 (kg/m ³)	0	0	0
α	Stainless steel	0	12.33×10^{-6} (1/K)	8.086×10^{-4}	0	0
	Zirconia	0	12.766×10^{-6} (1/K)	-1.491×10^{-3}	1.006×10^{-5}	-6.778×10^{-11}
k	Stainless steel	0	15.379 (W/mK)	-1.264×10^{-3}	2.092×10^{-6}	-7.233×10^{-10}
	Zirconia	0	1.700 (W/mK)	1.276×10^{-4}	6.648×10^{-8}	0
c	Stainless steel	0	496.56 (J/KgK)	-1.151×10^{-3}	1.636×10^{-6}	-5.863×10^{-10}
	Zirconia	0	487.34 (J/KgK)	3.049×10^{-4}	-6.037×10^{-8}	0

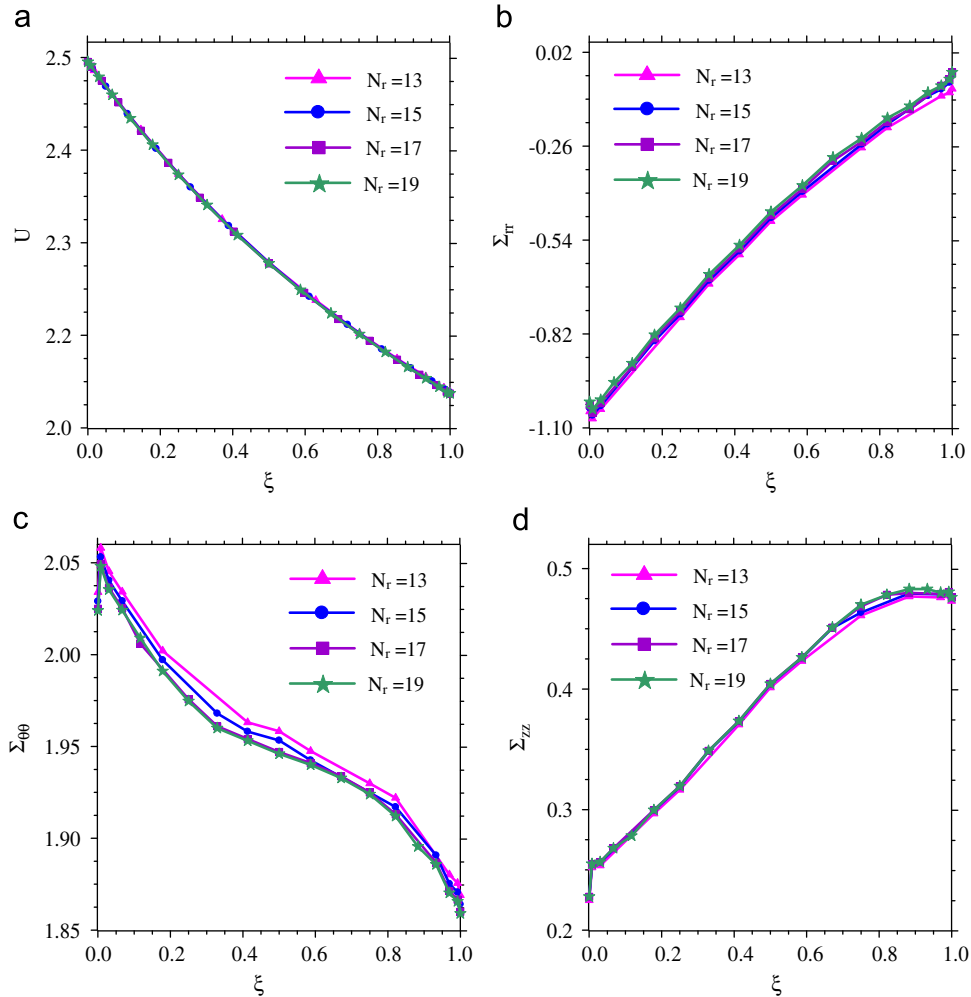


Fig. 2. The convergence of the results against the number of DQ grid points along the radial direction for the clamped FG cylindrical shells ($p=1, V_e=0.05, \bar{t}=0.5, N_e=24, \eta=0.5$).

Hyperbolic heat transfer equation:

$$\frac{1}{r} \frac{\partial}{\partial r} \left(rk \frac{\partial T}{\partial r} \right) + \frac{\partial}{\partial z} \left(k \frac{\partial T}{\partial z} \right) = \rho c \left(\frac{\partial T}{\partial t} + \tau \frac{\partial^2 T}{\partial t^2} \right) \quad (6)$$

Equations of motion:

$$\frac{\partial \sigma_{rr}}{\partial r} + \frac{\partial \sigma_{rz}}{\partial z} + \frac{\sigma_{rr} - \sigma_{\theta\theta}}{r} = \rho \frac{\partial^2 u}{\partial t^2} \quad (7)$$

$$\frac{\partial \sigma_{rz}}{\partial r} + \frac{\partial \sigma_{zz}}{\partial z} + \frac{\sigma_{rz}}{r} = \rho \frac{\partial^2 w}{\partial t^2} \quad (8)$$

where $k [=k(r,T)]$, $\rho [= \rho(r,T)]$ and $c [=c(r,T)]$ are the thermal conductivity, mass density and specific heat capacity at an arbitrary material point of the FG cylinder, respectively; also t is time and τ is the relaxation time.

Different thermal boundary conditions can be considered on the surfaces of the cylinder. Without loss of generality, the following boundary and initial conditions are assumed in this study,

At $z=0$: $T(r,z,t)=T_0$; at $z=L$: $T(r,z,t)=T_0(r,z)$

$$\text{or } \hat{q}_z = -k \frac{\partial T}{\partial z} = h_c(T-T_\infty) \quad (9a-c)$$

$$\text{At } r=R_i: -k \frac{\partial T}{\partial r} = \hat{q}_r = \begin{cases} q_0(t) & 0 \leq z \leq z_0(t) \\ 0 & z_0(t) \leq z \leq L \end{cases};$$

$$\text{at } r=R_o: \hat{q}_r = -k \frac{\partial T}{\partial r} = h_c(T-T_\infty) \quad (10a,b)$$

$$T(r,z,0)=T_0, \quad \left. \frac{\partial T}{\partial t} \right|_{t=0} = 0 \quad (11a,b)$$

where $T_0(r,z)$ is the initial temperature of the FG cylinder and T_∞ is the ambient temperature; $z_0(t)$ is the location of pressure and heat flux front edge; h_c is the convection heat transfer coefficient on the outer surface of the cylinder; $\hat{q}_n(n=r,z)$ and $q_0(t)$ are the

Table 2

Material properties of ceramic (Zirconia) and metal (steel) [5].

Material	Zirconia	Steel
k (W/cm °C)	2.09	20
ρ (kg/m ³)	5700	8166
c (J/kg °C)	531.9	325.35
E (MPa)	168.06	207.79
α (1/°C)	2.3×10^{-4}	1.5×10^{-5}
ν	0.298	0.3178

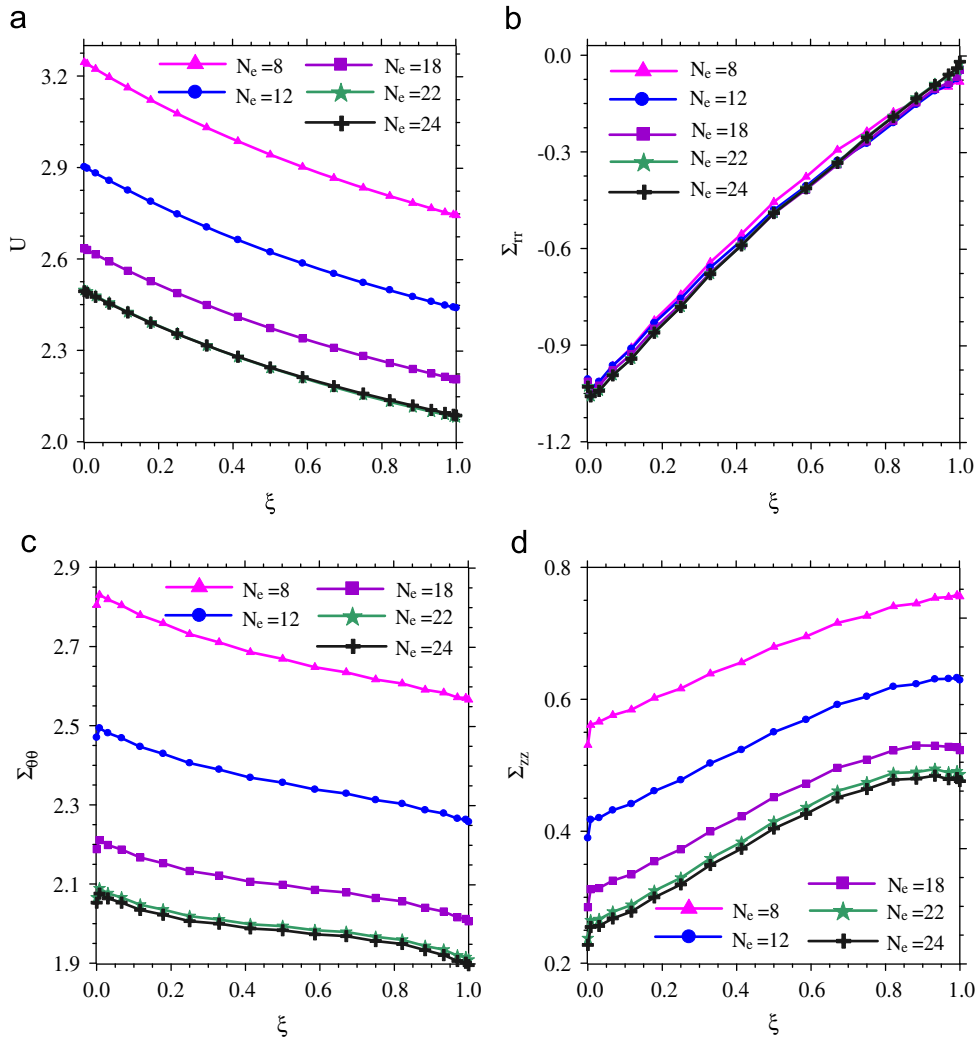


Fig. 3. The convergence of the results against the number of elements along the axial direction for the clamped FG cylindrical shells ($p=1$, $V_e=0.05$, $\bar{t}=0.5$, $N_r=19$, $\eta=0.5$).

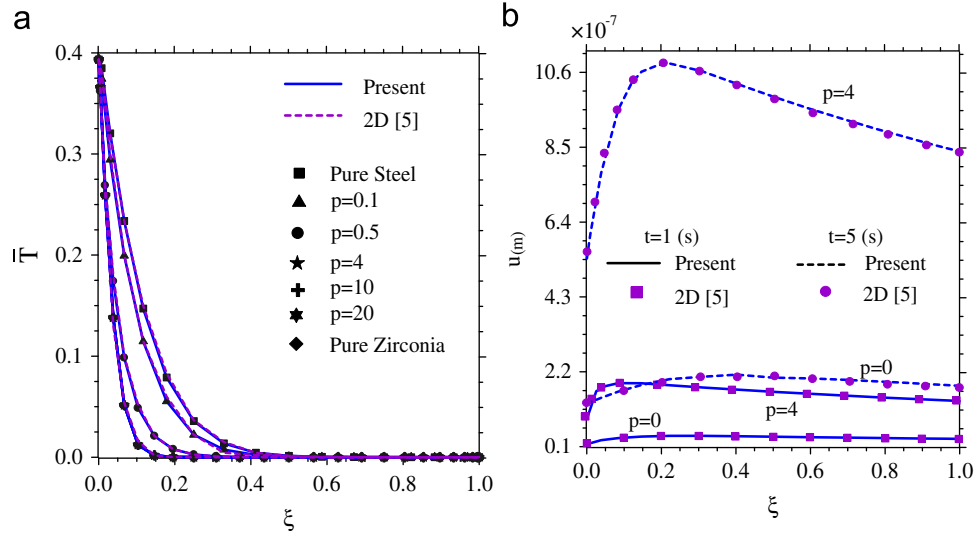


Fig. 4. Comparison of the transient non-dimensional temperature distributions and radial displacement component across the thickness of an FG hollow cylinder subjected to prescribed temperature on its inner surface [$R_i=4$ (cm), $R_o=6$ (cm), $L=20$ (cm), $\bar{T}=T/T_0$, $\eta=0.5$, $h_c=10$ (W/cm² °C)].

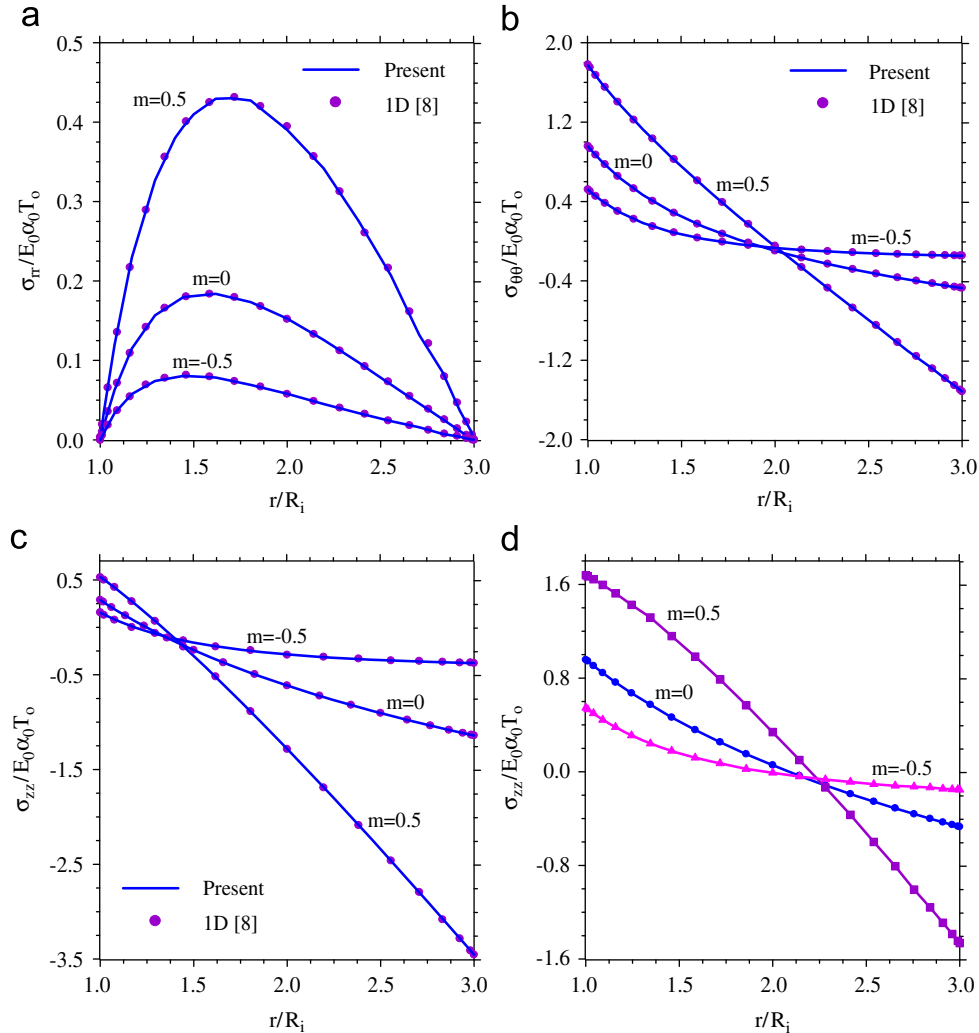


Fig. 5. Comparison of the stress components in the FG hollow cylinder subjected to non-uniform temperature rise with exact solution($R_o/R_i=3$, $L/R_i=10$, $R_i=1$, $\eta=0.5$).

prescribed heat flux on the surface boundaries and the heat source strength, respectively.

The mechanical boundary conditions on the inner and outer surfaces of the cylinder are

$$\text{At } r = R_i : \hat{\sigma}_{rz} = 0, \hat{\sigma}_{rr} = \begin{cases} -P_0(t) & 0 \leq z \leq z_0(t) \\ 0 & z_0(t) \leq z \leq L \end{cases};$$

$$\text{at } r = R_o : \hat{\sigma}_{rr} = \hat{\sigma}_{rz} = 0 \quad (12a-d)$$

Table 3

Comparison of the results for the FG cylindrical shell subjected to dynamic pressure [$R_o=0.1$ (m), $R_i=0.08$ (m), $L=1$ (m), $\eta=0.5$].

Source	t^*	U^*	$\Sigma_{\theta\theta}^*$	Σ_{zz}^*	Σ_{rr}^*
Present	0.1	0.202	1.102	0.238	-0.309
ANSYS [9]		0.201	1.109	0.232	-0.296
2D [9]		0.196	1.082	0.237	-0.290
Present	0.3	0.530	2.855	0.607	-0.809
ANSYS [9]		0.525	2.903	0.606	-0.774
2D [9]		0.505	2.783	0.611	-0.748
Present	0.5	0.655	3.566	0.764	-1.000
ANSYS [9]		0.652	3.589	0.749	-0.956
2D [9]		0.637	3.510	0.771	-0.945

where $\hat{\sigma}_{ij}$ ($i, j=r, z$) are the prescribed stress tensor components on the boundaries of the cylinder. Different types of boundary conditions at the edges $z=0$ and L can be considered which can be obtained from the following conditions:

$$\text{Either } u(r, z, t) = 0 \text{ or } \hat{\sigma}_{rz} = 0; \text{ either } w(r, z, t) = 0 \text{ or } \hat{\sigma}_{zz} = 0 \quad (13a-d)$$

For example, for the FG cylinder with clamped and free edges one has,

$$\begin{aligned} \text{Clamped edge : } u(r, z, t) = 0, \quad w(r, z, t) = 0; \\ \text{free edge : } \hat{\sigma}_{rz} = 0, \quad \hat{\sigma}_{zz} = 0 \end{aligned} \quad (14a-d)$$

It is assumed that before any loading, the FG cylinder is at rest in the undeformed configuration. Hence,

$$u(r, z, 0) = w(r, z, 0) = 0, \quad \left. \frac{\partial u}{\partial t} \right|_{t=0} = \left. \frac{\partial w}{\partial t} \right|_{t=0} = 0 \quad (15a-d)$$

3. Solution procedure

Due to complex thermal and mechanical boundary conditions on the inner surface of the cylinder, the finite element method is employed to discretize the governing equations along the axial direction (z -axis). For this purpose, firstly, the weak form of Eqs. (6)–(8) are obtained as

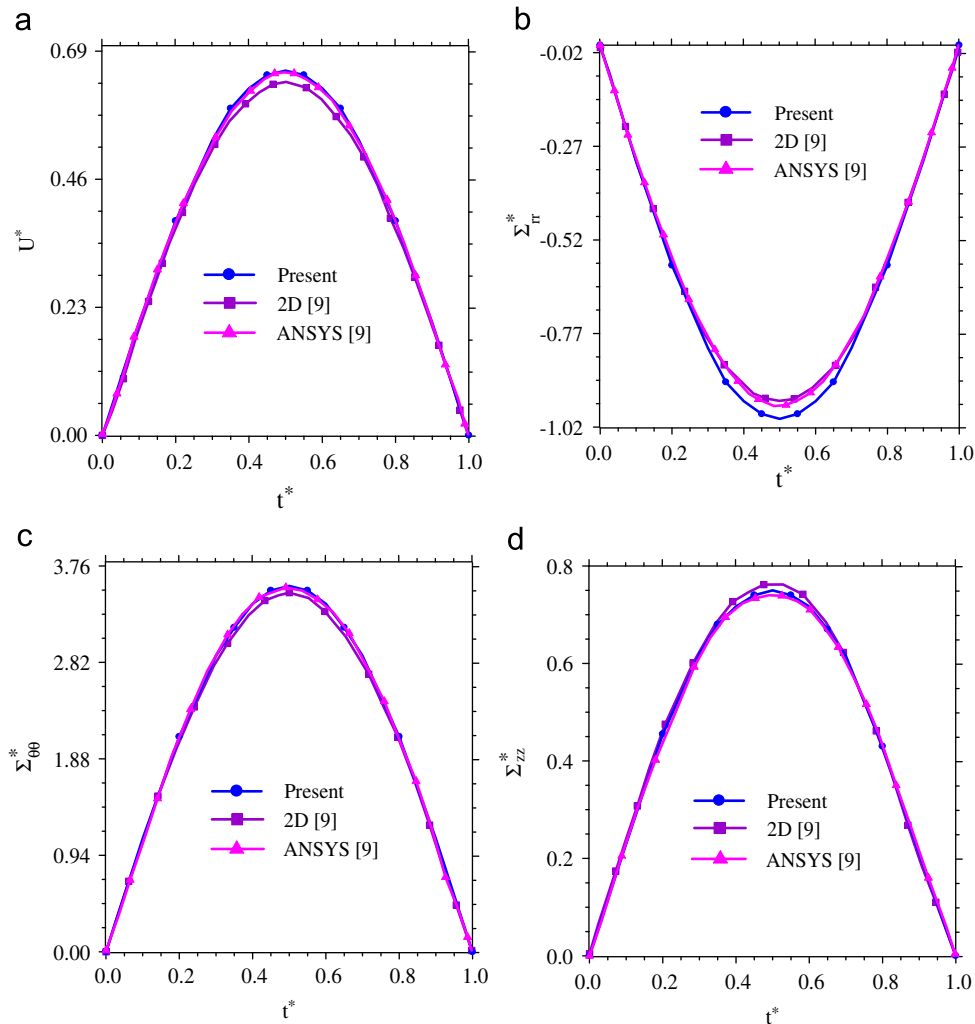


Fig. 6. Comparison of the stress and displacement components across the thickness of the FG cylindrical shell subjected to dynamic pressure [$R_o=0.1$ (m), $R_i=0.08$ (m), $L=1$ (m), $\eta=0.5$].

Eq. (6):

$$\int_0^L \int_{R_i}^{R_o} \left[k \frac{\partial(\delta T)}{\partial r} \frac{\partial T}{\partial r} + k \frac{\partial(\delta T)}{\partial z} \frac{\partial T}{\partial z} + \rho c \left(\frac{\partial T}{\partial t} + \tau \frac{\partial^2 T}{\partial t^2} \right) \delta T \right] r dr dz + \int_0^L (r \hat{q}_r \delta T) \Big|_{r=R_i}^{R_o} dz + \int_{R_i}^{R_o} (\hat{q}_z \delta T) \Big|_{z=0}^L r dr = 0 \quad (16)$$

Eq. (7):

$$\int_0^L \int_{R_i}^{R_o} \left[\frac{\partial(r \delta u)}{\partial r} \sigma_{rr} + \frac{\partial(r \delta u)}{\partial z} \sigma_{rz} - (\sigma_{rr} - \sigma_{\theta\theta}) \delta u + \rho r \frac{\partial^2 u}{\partial t^2} \delta u \right] r dr dz - \int_0^L (r \hat{\sigma}_{rr} \delta u) \Big|_{r=R_i}^{R_o} dz - \int_{R_i}^{R_o} (\hat{\sigma}_{rz} \delta u) \Big|_{z=0}^L r dr = 0 \quad (17)$$

Eq. (8):

$$\int_0^L \int_{R_i}^{R_o} \left[\frac{\partial(r \delta w)}{\partial z} \sigma_{zz} + \frac{\partial(r \delta w)}{\partial r} \sigma_{rz} - \sigma_{rz} \delta w + \rho r \frac{\partial^2 w}{\partial t^2} \delta w \right] r dr dz - \int_0^L (r \hat{\sigma}_{rz} \delta w) \Big|_{r=R_i}^{R_o} dz - \int_{R_i}^{R_o} (\hat{\sigma}_{zz} \delta w) \Big|_{z=0}^L r dr = 0 \quad (18)$$

where $\delta T, \delta u$ and δw are the variation of temperature and displacement components along the r and z -directions, respectively.

The FG cylinder is divided into N_e elements along the z -direction. In each element, the weak form of the governing equations, i.e. Eqs. (16)–(18), should be satisfied. In order to avoid the assembling procedure and consequently to reduce the

computational effort of the FEM procedure, the global interpolation functions are used to approximate the temperature distribution and displacement components within each element

$$T(r, z, t) = \sum_{i=1}^{N_z} \bar{T}_i(r, t) \phi_i(z), \quad u(r, z, t) = \sum_{i=1}^{N_z} U_i(r, t) \phi_i(z),$$

$$w(r, z, t) = \sum_{i=1}^{N_z} W_i(r, t) \phi_i(z) \quad (19a-c)$$

Where N_z is the total number of nodes along the z -direction and $\phi_i(z)$ ($i=1, 2, \dots, N_z$) are the global Lagrange interpolation functions, which for linear elements take the following form:

$$\phi_i(z) = \begin{cases} 0 & z \leq z_{i-1} \\ \frac{z - z_{i-1}}{z_i - z_{i-1}} & z_{i-1} \leq z \leq z_i \\ \frac{z_{i+1} - z}{z_{i+1} - z_i} & z_i \leq z \leq z_{i+1} \\ 0 & z_{i+1} \leq z \end{cases} \quad \text{for } i = 1, 2, \dots, N_z \quad (20)$$

Substituting the approximation presented in Eqs. (19) into Eqs. (16)–(18) and evaluating the integration over the length of the cylinder, one gets

$$\int_{R_i}^{R_o} \left[\sum_{i=1}^{N_z} \sum_{j=1}^{N_z} \left[-k_{ij} \left(\frac{\partial^2 \bar{T}_j}{\partial r^2} + \frac{1}{r} \frac{\partial \bar{T}_j}{\partial r} \right) - \frac{\partial k_{ij}}{\partial r} \frac{\partial \bar{T}_j}{\partial r} + k'_{ij} \bar{T}_j \right] \right]$$

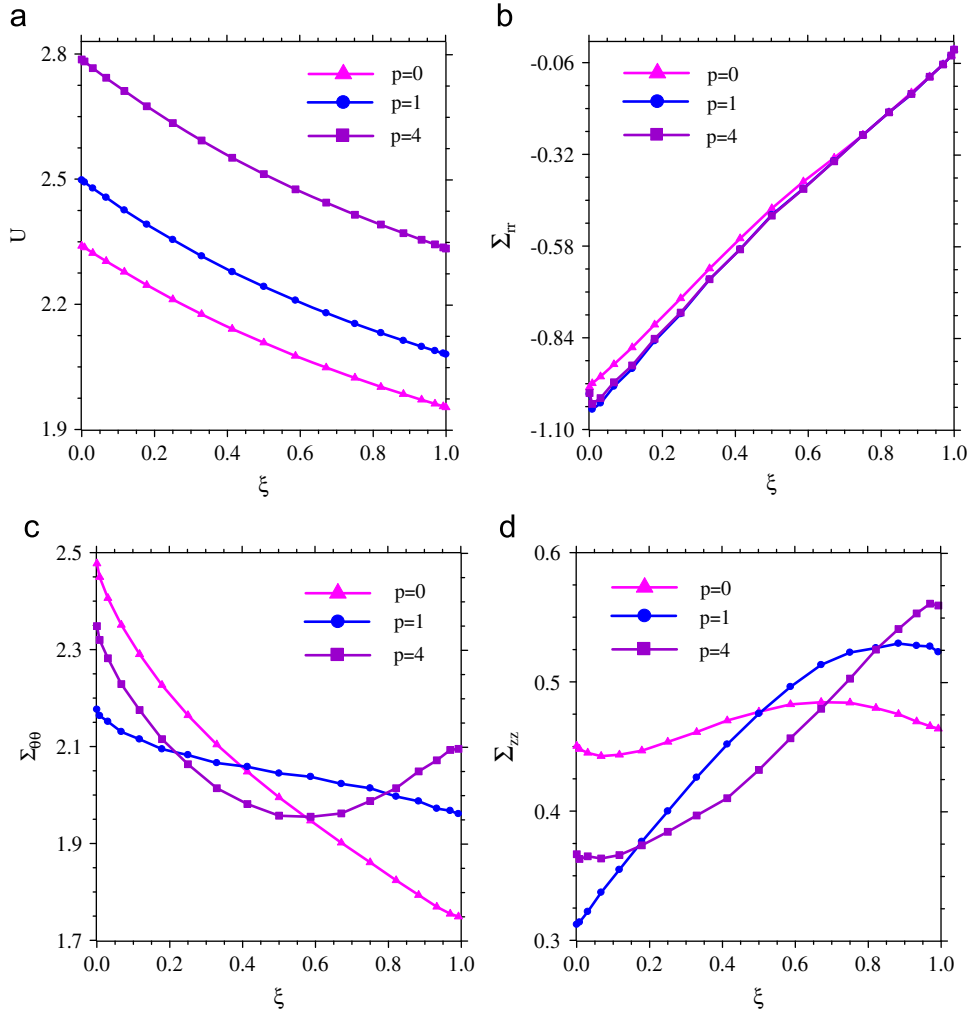


Fig. 7. Influence of the material graded index p on the results for the clamped FG cylindrical shells ($V_e=0.05$, $\bar{t}=0.5$, $\eta=0.5$).

$$+ H_{ij} \frac{\partial \bar{T}_j}{\partial t} + \bar{H}_{ij} \frac{\partial^2 \bar{T}_j}{\partial t^2} \delta \bar{T}_i r dr - \left\{ r \sum_{i=1}^{N_z} \left[\int_0^L \phi_i \hat{q}_r dz + \sum_{j=1}^{N_z} k_{ij} \frac{\partial \bar{T}_j}{\partial r} \right] \delta \bar{T}_i \right\} \Big|_{r=R_i}^{R_o} - \int_{R_i}^{R_o} \left(\sum_{i=1}^{N_z} \phi_i \hat{q}_z \delta \bar{T}_i \right) \Big|_{z=0}^L r dr = 0 \quad (21)$$

$$\int_{R_i}^{R_o} \sum_{i=1}^{N_z} \left[\sum_{j=1}^{N_z} \left\{ -A_{ij}^{ij} \left(\frac{\partial^2 U_j}{\partial r^2} + \frac{1}{r} \frac{\partial U_j}{\partial r} \right) - \frac{\partial A_{11}^{ij}}{\partial r} \frac{\partial U_j}{\partial r} - \frac{1}{r} \left(\frac{\partial A_{12}^{ij}}{\partial r} - \frac{A_{22}^{ij}}{r} \right) U_j - B_{12}^{ij} \frac{\partial W_j}{\partial r} - \frac{\partial B_{12}^{ij}}{\partial r} W_j + B_{55}^{ij} \frac{\partial W_j}{\partial r} + D_{55}^{ij} U_j + M_{ij} \frac{\partial^2 U_j}{\partial t^2} \right\} + \frac{\partial \Theta_i}{\partial r} \right] \delta U_i r dr - \left\{ r \sum_{i=1}^{N_z} \left[\int_0^L \phi_i \hat{\sigma}_{rr} dz + \sum_{j=1}^{N_z} \left(A_{11}^{ij} \frac{\partial U_j}{\partial r} + \frac{A_{12}^{ij}}{r} U_j + B_{12}^{ij} W_j \right) - \Theta_i \right] \delta U_i \right\} \Big|_{r=R_i}^{R_o} - \int_{R_i}^{R_o} \left(\sum_{i=1}^{N_z} \phi_i \hat{\sigma}_{rz} \delta U_i \right) \Big|_{z=0}^L r dr = 0 \quad (22)$$

$$\int_{R_i}^{R_o} \sum_{i=1}^{N_z} \left[\sum_{j=1}^{N_z} \left\{ -A_{55}^{ij} \left(\frac{\partial^2 W_j}{\partial r^2} + \frac{1}{r} \frac{\partial W_j}{\partial r} \right) - \frac{\partial A_{55}^{ij}}{\partial r} \frac{\partial W_j}{\partial r} - B_{55}^{ij} \left(\frac{\partial U_j}{\partial r} + \frac{U_j}{r} \right) - \frac{\partial B_{55}^{ij}}{\partial r} U_j + B_{12}^{ij} \left(\frac{\partial U_j}{\partial r} + \frac{U_j}{r} \right) + D_{11}^{ij} W_j + M_{ij} \frac{\partial^2 W_j}{\partial t^2} \right\} - \Theta'_i \right] \delta W_i r dr$$

$$- \left\{ r \sum_{i=1}^{N_z} \left[\int_0^L \phi_i \hat{\sigma}_{rz} dz + \sum_{j=1}^{N_z} \left(A_{55}^{ij} \frac{\partial W_j}{\partial r} + B_{55}^{ij} U_j \right) \right] \delta W_i \right\} \Big|_{r=R_i}^{R_o} dz - \int_{R_i}^{R_o} \left(\sum_{i=1}^{N_z} \phi_i \hat{\sigma}_{zz} \delta W_i \right) \Big|_{z=0}^L r dr = 0 \quad (23)$$

where

$$A_{ij}^{rs} = \int_0^L C_{ij} \phi_r \phi_s dz, \quad B_{ij}^{rs} = \int_0^L C_{ij} \phi_r \frac{d\phi_s}{dz} dz,$$

$$D_{ij}^{rs} = \int_0^L C_{ij} \frac{d\phi_r}{dz} \frac{d\phi_s}{dz} dz, \quad H_{ij} = \int_0^L \phi_i \phi_j \rho c dz,$$

$$\bar{H}_{ij} = \int_0^L \phi_i \phi_j \tau \rho c dz, \quad M_{ij} = \int_0^L \phi_i \phi_j \rho dz,$$

$$\Theta_i(r) = \int_0^L \phi_i (C_{11} + 2C_{12}) \varepsilon_T(r, z) dz,$$

$$\Theta'_i = \int_0^L (C_{11} + 2C_{12}) \varepsilon_T(r, z) \frac{d\phi_i}{dz} dz \quad (24a-h)$$

Using the linear independence of the terms $\delta \bar{T}_i$, δU_i and δW_i ($i=1, 2, \dots, N_z$) and after implementing the boundary conditions at the surfaces $z=0$ and L , the governing differential equation and the related boundary conditions at each FE nodal line 'i' with

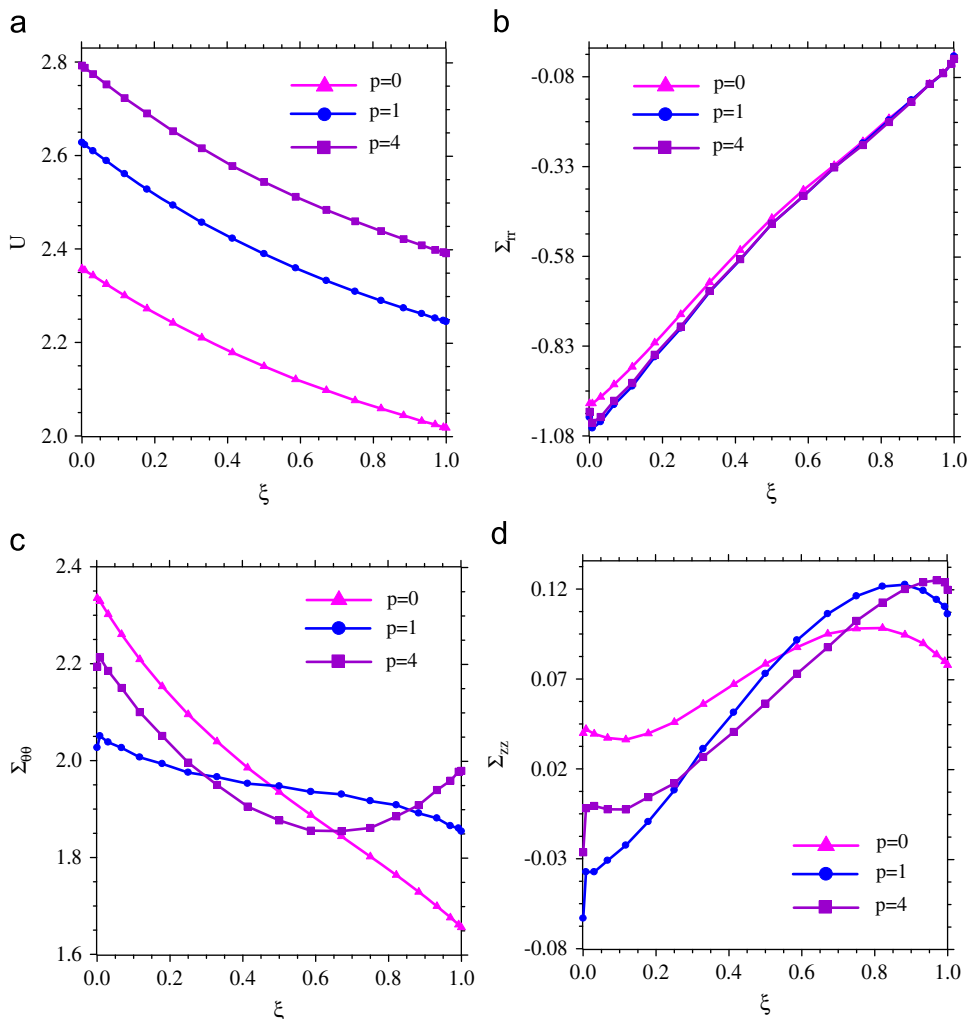


Fig. 8. Influence of the material graded index p on the results for the clamped-free FG cylindrical shells ($V_c=0.05$, $\bar{t}=0.5$, $\eta=0.5$).

$i = 1, 2, \dots, N_z$ are obtained as follows:

$$\sum_{j=1}^{N_z} \left[-k_{ij} \left(\frac{\partial^2 \bar{T}_j}{\partial r^2} + \frac{1}{r} \frac{\partial \bar{T}_j}{\partial r} \right) - \frac{\partial k_{ij}}{\partial r} \frac{\partial \bar{T}_j}{\partial r} + k_{ij}'' \bar{T}_j + H_{ij} \frac{\partial \bar{T}_j}{\partial t} + \bar{H}_{ij} \frac{\partial^2 \bar{T}_j}{\partial t^2} \right] = 0 \quad (25)$$

$$\sum_{j=1}^{N_z} \left\{ -A_{11}^{ij} \left(\frac{\partial^2 U_j}{\partial r^2} + \frac{1}{r} \frac{\partial U_j}{\partial r} \right) - \frac{\partial A_{11}^{ij}}{\partial r} \frac{\partial U_j}{\partial r} - \frac{1}{r} \left(\frac{\partial A_{12}^{ij}}{\partial r} - \frac{A_{22}^{ij}}{r} \right) U_j - B_{12}^{ij} \frac{\partial W_j}{\partial r} - \frac{\partial B_{12}^{ij}}{\partial r} W_j + B_{55}^{ij} \frac{\partial W_j}{\partial r} + D_{55}^{ij} U_j + M_{ij} \frac{\partial^2 U_j}{\partial t^2} \right\} = -\frac{\partial \Theta_i}{\partial r} \quad (26)$$

$$\sum_{j=1}^{N_z} \left\{ -A_{55}^{ij} \left(\frac{\partial^2 W_j}{\partial r^2} + \frac{1}{r} \frac{\partial W_j}{\partial r} \right) - \frac{\partial A_{55}^{ij}}{\partial r} \frac{\partial W_j}{\partial r} - B_{55}^{ij} \left(\frac{\partial U_j}{\partial r} + \frac{U_j}{r} \right) - \frac{\partial B_{55}^{ij}}{\partial r} U_j + B_{12}^{ij} \left(\frac{\partial U_j}{\partial r} + \frac{U_j}{r} \right) + D_{11}^{ij} W_j + M_{ij} \frac{\partial^2 W_j}{\partial t^2} \right\} = \Theta_i' \quad (27)$$

Thermal boundary conditions:

$$\text{At } r = R_i : \sum_{j=1}^{N_z} k_{ij} \frac{\partial \bar{T}_j}{\partial r} = f_i(t);$$

$$\text{at } r = R_o : \sum_{j=1}^{N_z} \left(k_{ij} \frac{\partial \bar{T}_j}{\partial r} + \hat{A}_{ij} \bar{T}_j \right) = g_i(t) \quad (28a, b)$$

where

$$f_i(t) = - \int_0^L \phi_i \hat{q}_r dz, \quad \hat{A}_{ij} = \int_0^L h_c \phi_i \phi_j dz, \quad g_i(t) = \int_0^L h_c T_\infty \phi_i \phi_j dz \quad (29a-c)$$

Mechanical boundary conditions:

$$\text{At } r = R_i, R_o : \sum_{j=1}^{N_z} \left(A_{11}^{ij} \frac{\partial U_j}{\partial r} + A_{12}^{ij} \frac{U_j}{r} + B_{12}^{ij} W_j \right) - \Theta_i = P(t),$$

$$\sum_{j=1}^{N_z} \left(A_{55}^{ij} \frac{\partial W_j}{\partial r} + B_{55}^{ij} U_j \right) = 0 \quad (30a, b)$$

where

$$P_i(t) = - \int_0^L \phi_i \hat{\sigma}_{rr} dz \quad (31)$$

It is obvious that the resulting governing differential equations have variable coefficients and consequently if it is not impossible to solve them analytically, it is very difficult to obtain such a solution. Hence, here the differential quadrature method (DQM) as an efficient and accurate numerical tool [9,10,20–23] in conjunction with Newmark time integration scheme are employed to discretize this system of equations in the radial direction and temporal domain, respectively. A brief review of DQM is presented in [Appendix A](#). Based on the DQM, at each finite element nodal line 'i', the cylinder is

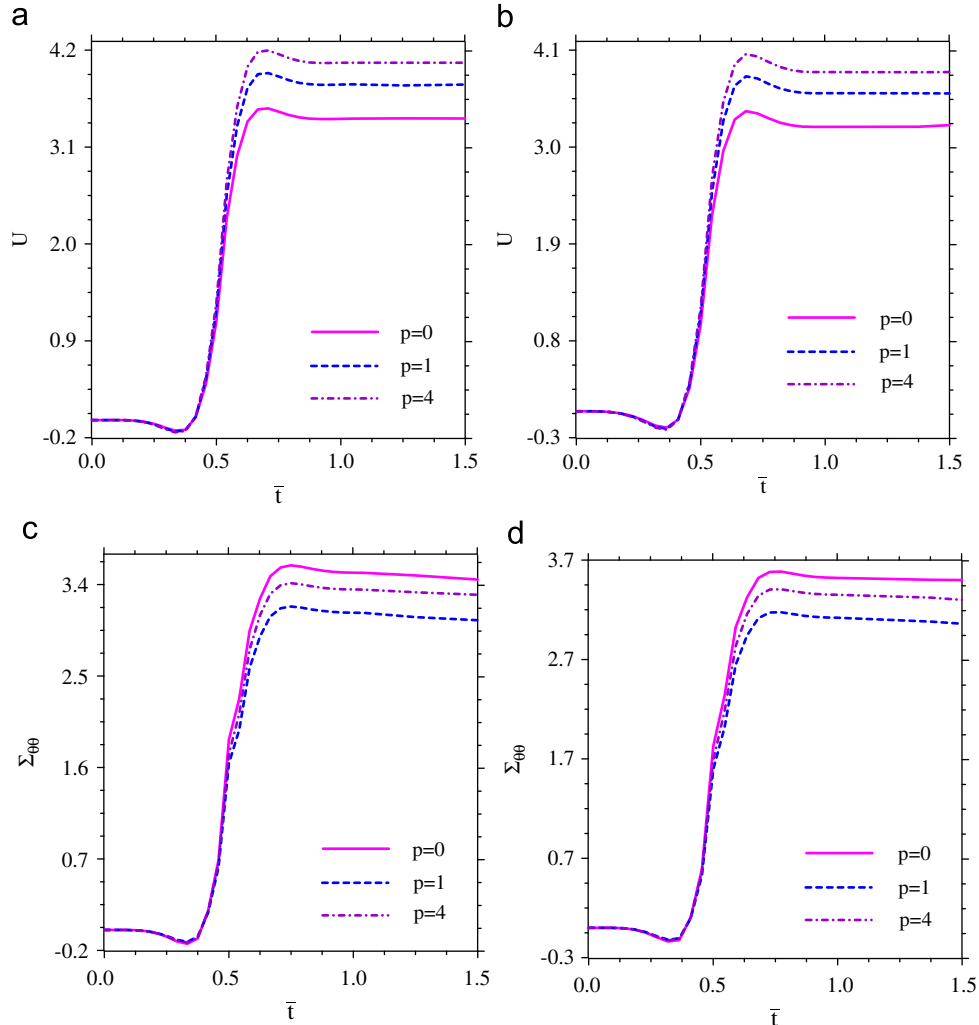


Fig. 9. Influence of the material graded index p on the time history of the results at the point $(\xi=0, \eta=0.5)$ for the FG cylindrical shells ($V_e=0.05$).

discretized into a set of N_r discrete grid points along the r -direction. Then, at each domain grid point $r_j (j=2, 3, \dots, N_r-1)$, the governing differential equations and at the boundary grid points $r_j (j=1, N_r)$, the boundary conditions are discretized. Here, for brevity purpose, only the DQ discretized of the heat conduction Eq. (25) is represented,

$$\sum_{m=1}^{N_z} \left\{ - \sum_{n=1}^{N_r} \left[\left(\frac{k_{im}}{r} + \frac{\partial k_{im}}{\partial r} \right) A_{jn}^r + (k_{im})_j B_{jn}^r \right] \bar{T}_{mn} + (k_{im}^r)_j \bar{T}_{mj} + (H_{im})_j \frac{d\bar{T}_{mj}}{dt} + (H_{im})_j \frac{d^2 \bar{T}_{mj}}{dt^2} \right\} = 0 \quad (32)$$

where A_{ij}^r and B_{ij}^r represent the weighting coefficients of the first and second-order derivatives along the r -direction, respectively [23]; also $(\cdot)_j$ means the function value at $r=r_j$. After doing these, the discretized governing differential equations and the associated boundary conditions can be represented as in the matrix form as

Heat transfer equations:

$$[\bar{M}] \left\{ \frac{d^2 \bar{T}}{dt^2} \right\} + [\bar{C}] \left\{ \frac{d\bar{T}}{dt} \right\} + [\bar{K}] \{\bar{T}\} = \{\bar{F}\} \quad (33)$$

Equations of motion:

$$[\hat{M}] \left\{ \frac{d^2 D}{dt^2} \right\} + [\hat{K}] \{D\} = \{\hat{F}(\bar{T}, T_0, t)\} \quad (34)$$

where $\bar{T}$ and $\{D\}$ are the temperature and displacement components vector of degrees of freedom. Based on the arrangement of elements these vectors, the elements of the coefficient matrixes $[\bar{M}]$, $[\bar{C}]$, $[\bar{K}]$, $[\hat{M}]$, $[\hat{K}]$ and the load vectors $\bar{F}$ and $\hat{F}$ are obtained from the discretized form of the governing differential equations and the related boundary conditions.

Using the Newmark time integration scheme [24], firstly, the system of differential Eq. (33) subjected to the related initial conditions is solved in each time step iteratively. Then, the evaluated temperature distribution is used in Eq. (34). After solving this system of equation, the displacement components at that time step and at all spatial grid points are obtained. Then, using the constitutive relations, the stress tensor components at any point of the FG shell can be evaluated.

4. Numerical results

In this section, firstly, the convergence behavior of the method for evaluating the non-dimensional displacement and stress components across the thickness of an FG hollow cylinder under moving boundary pressure and heat flux is investigated. Without loss of generality, in all problems considered here, equal length elements are used. The FG material properties are assumed to

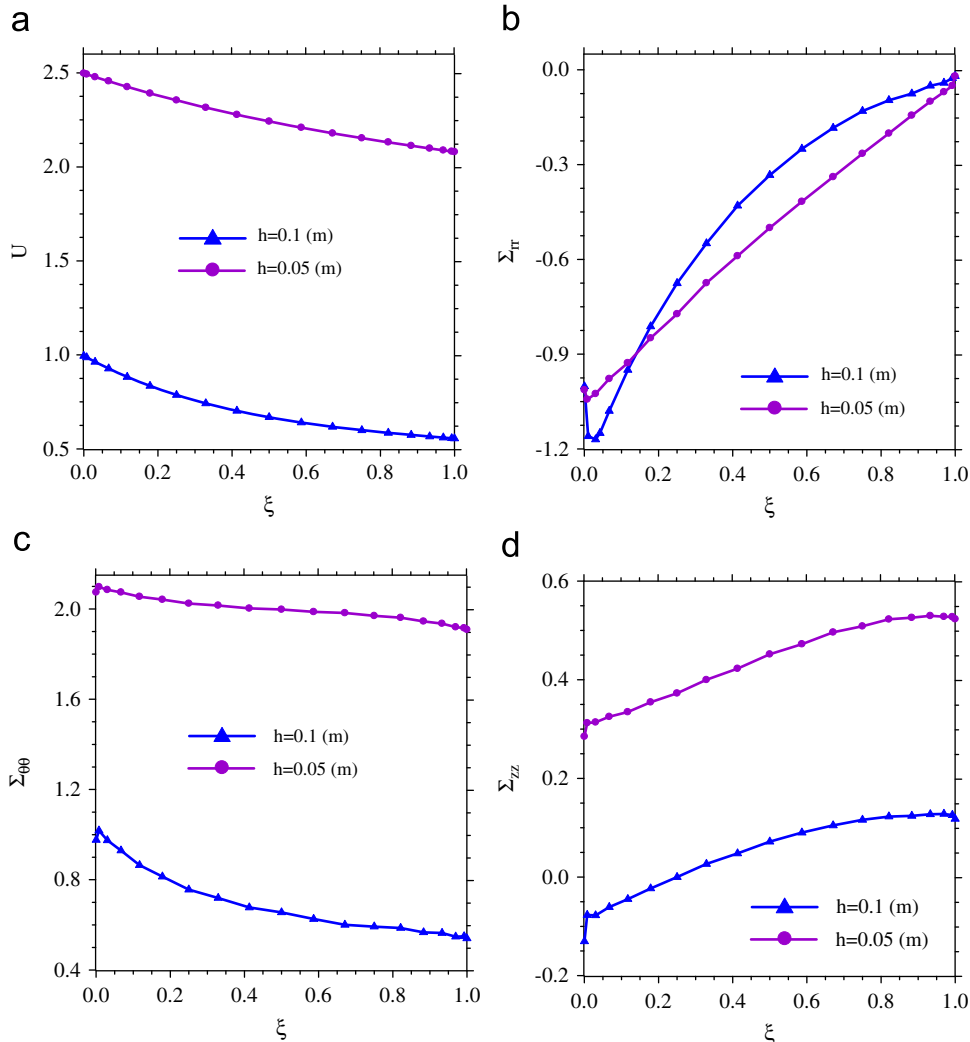


Fig. 10. Comparison of the results for the clamped FG cylindrical shells with two different thickness ($p=1$, $V_e=0.05$, $\bar{T}=0.5$, $\eta=0.5$).

vary according to power law distribution (1); also, the non-dimensional parameters are defined as

$$\eta = z/L, \quad \xi = \frac{r-R_i}{R_o-R_i}, \quad U = \frac{uE_{0c}}{R_oP_0}, \quad \Sigma_{ii} = \frac{\sigma_{ii}}{P_0} \quad (i=r, \theta, z),$$

$$\bar{t} = \frac{\nu_0 t}{L}, \quad V_e = \sqrt{\frac{\hat{\alpha}\tau}{h^2}} \quad (35a-f)$$

where $\hat{\alpha}(=k/\rho c)$, $h(=R_o-R_i)$ and L are the thermal diffusivity, thickness and length of the FG hollow cylinder, respectively; ν_0 is the velocity of the pressure and heat flux front edge, which is assumed to be constant. The effect of relaxation time τ is appeared in Vernotte number (V_e) and it can be seen from Eq. (6) that by increasing the Vernotte number, the non-Fourier effect increases. Also, it is obvious that when $V_e=0$, the results become equal to those obtained using Fourier model (please see Eq. (6)).

Otherwise specified, the following values for the different physical and geometrical parameters are used in the solved examples,

$$\nu_0 = 0.25 \text{ (m/s)}, \quad q_0 = 5 \text{ (KW/m}^2\text{)}, \quad P_0 = 100 \text{ (MPa)},$$

$$h_c = 10 \text{ (W/m}^2 \text{ } ^\circ\text{K)}, \quad R_i = 0.15 \text{ (m)}, \quad R_o = 0.2 \text{ (m)}, \quad L = 1 \text{ (m)} \quad (36)$$

Also, the material properties of Zirconia (ceramic) and steel (metal) as given in Table 1, which are chosen from the work of Santos et al. [5], are used to obtain the new numerical results.

As a first example, the convergence behavior of the non-dimensional radial displacement and stress components across the thickness of an FG cylinder against the number of DQ grid points along the r -direction and the number of elements along the z -direction are shown in Figs. 2 and 3, respectively. It is noticeable that increasing the number of DQ grid points and elements, the results converges without any numerical instability. Based on this study, it is found that a set of $N_r=19$ DQ grid points along the radial direction is sufficient regardless of the end boundary conditions of the FG cylinder. But, the number of element along the axial direction slightly depends on the end boundary conditions; for example a number of $N_e=22$ and $N_e=24$ should be used for the dynamic analysis of clamped-clamped and clamped-free FG hollow cylinder, respectively.

In order to verify the presented approach for the transient analysis of FG hollow cylinder under thermal load, an FG thick-walled hollow cylinder subjected to a prescribed transient temperature on its inner surface studied by Santos et al. [5] using a semi-analytical finite element model is analyzed here. Completely free FG hollow cylinder with zero initial displacement and velocity components were considered by them. The thermal conditions at the different surface boundaries of the FG hollow cylinder and also, the initial thermal conditions are as follows [5]:

$$\text{At } r = R_i : T(r, z, t) = T_0(1 - e^{-0.5t});$$

$$\text{at } r = R_o : k \frac{\partial T}{\partial r} + h_c T = 0; \quad \text{at } z = 0, L : T(r, z, t) = 0$$

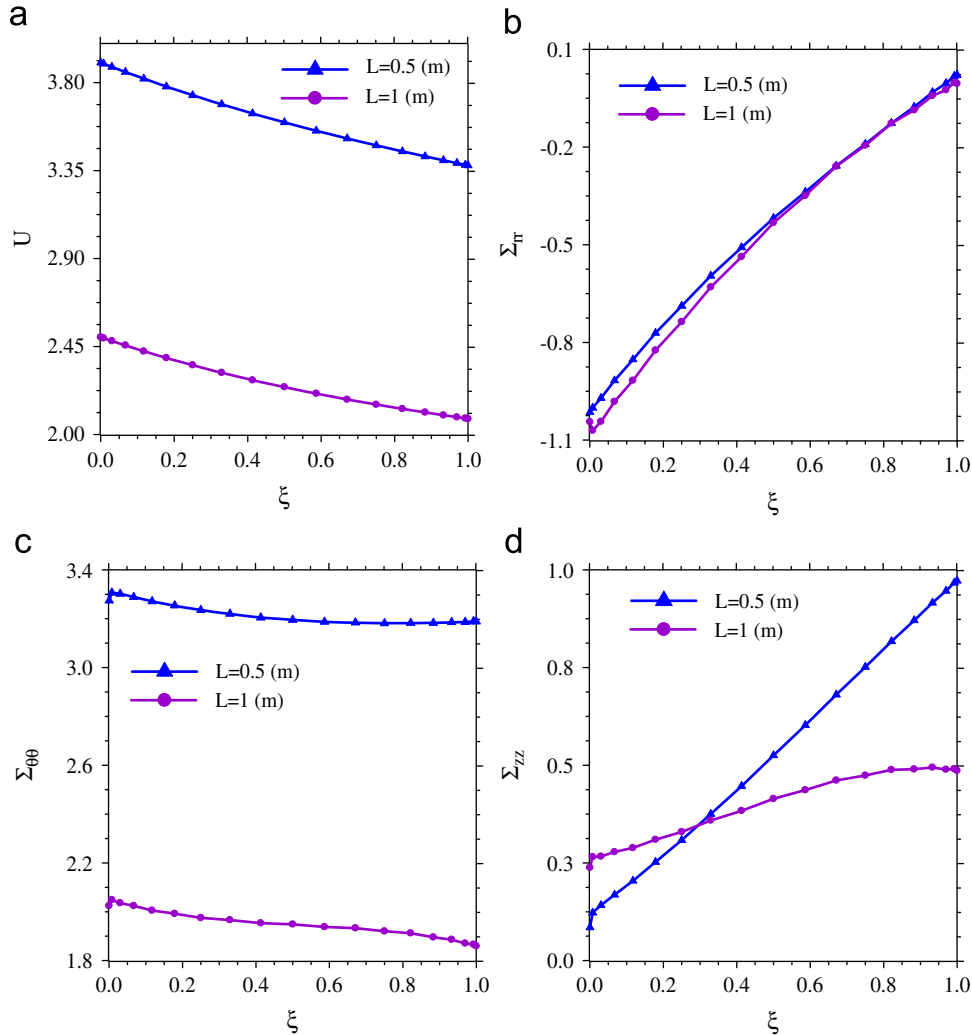


Fig. 11. Comparison of the results for the clamped FG cylindrical shells with two different length ($p=1$, $V_e=0.05$, $t=2$ (s), $\eta=0.5$).

$$T(r, z, 0) = 0, \quad \left. \frac{\partial T(r, z, t)}{\partial t} \right|_{t=0} = 0 \quad (37a-e)$$

The material properties are given in Table 2. The temperature distribution and the radial displacement component across the thickness of the FG hollow cylinder for different values of the material graded index (p) are compared with those of Santos et al. [5] in Fig. 4. Excellent agreement between the results of the two approaches is evident.

As another comparison study for thermoelastic analysis, a completely free long FG cylindrical shell subjected to a thermal loading which has analytical solution [8] is considered. Peng and Li [8] transformed the one-dimensional thermoelasticity equation into a Fredholm integral equation to obtain an analytical solution. To find such a solution, they assumed that all the material properties, except Poisson's ratio, vary according to $\psi = \psi_0 r^m$ where ψ_0 is a material constant at the outer surface of the cylinder and m is the material graded index. The material properties are as follows [8]:

$$E_0 = 200 \text{ GPa}, \quad \nu_0 = 0.3, \quad \alpha_0 = 1.2 \times 10^{-6} \text{ (1/}^\circ\text{C)} \quad (38)$$

Also the surface temperature at the inner and outer surfaces of FG cylinder is assumed to be

$$T(R_i) = 0^\circ\text{C}, \quad T(R_o) = 100^\circ\text{C} \quad (39a, b)$$

The non-dimensional stress components at the middle section of the FG cylinder are compared with those of Peng and Li [8] for different values of the material graded index (m) in Fig. 5. Again, one can see that excellent agreement between the results of the two approaches exists. It should be noted that Peng and Li [8] employed the plane strain assumption to obtain the axial stress component,

$$\sigma_{zz} = \nu(\sigma_{rr} + \sigma_{\theta\theta}) \quad (40)$$

In this study, in order to compare the results with those of them, this component of stress is evaluated using Eq. (40). However, the results without considering this assumption are also evaluated and are shown in Fig. 5(d).

In order to further verify the presented approach, FG hollow cylinders subjected to internal dynamic pressure as a limiting case of FG hollow cylinders subjected to an axially moving boundary pressure is analyzed here. For this purpose, the problem considered by Setoodeh et al. [9] is adopted. They obtained the dynamic response of FG hollow cylinders under a uniformly distributed but transient pressure $P(t) = P_0 \sin(\pi t)$ on its inner surface. The outer surface was assumed to be stress free. They employed the layerwise finite element through the thickness of the shell and DQM along the axial direction (LWDQ). In addition, they modeled the problem using the commercial finite element software ANSYS. One-quadrant of the shell structure was simulated using 18,000 3D solid elements with

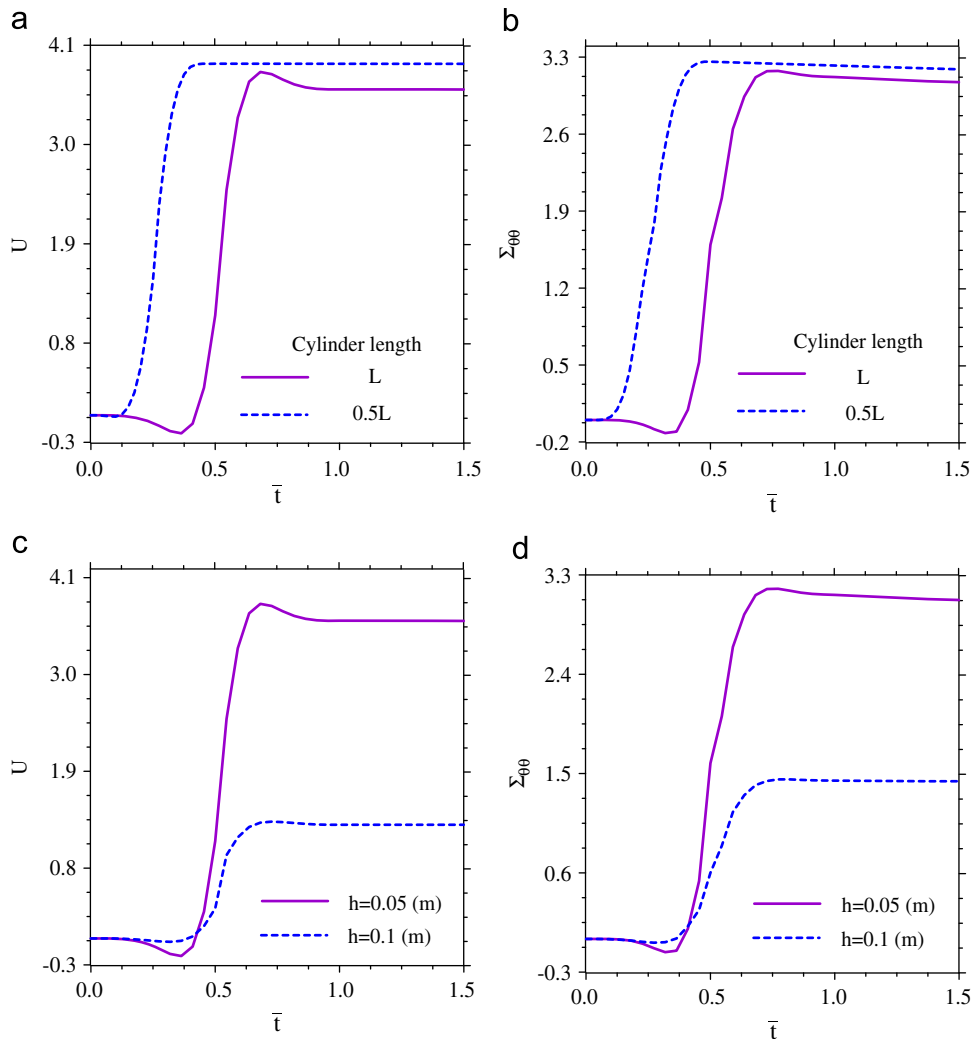


Fig. 12. Influence of the length and thickness on the time history of the results at the point ($\xi=0, \eta=0.5$) for the clamped FG cylindrical shells [$V_e=0.05, p=1, \nu_0=0.25$ (m/s), $L=1$ (m), $\bar{t}=(v_0 t)/L$].

20 nodes. They assumed that the modulus of elasticity and the mass density varies through the shell thickness according to, respectively,

$$E(r) = E_m \left(\frac{r}{R_i} \right)^{n_1}, \quad \rho(r) = \rho_m \left(\frac{r}{R_i} \right)^{n_2} \quad (41a, b)$$

where E_m and ρ_m are the metal modulus of elasticity and density, respectively. n_1 and n_2 are the material graded index. The material properties and also the non-dimensional parameters of the FG shell are as follows [9]:

$$E_m = 223 \text{ GPa}, \quad \rho_m = 8900 \text{ (Kg/m}^3\text{)}, \quad \nu = 0.3, \quad n_1 = 2, \\ n_2 = -5.93, \quad t^* = \frac{t}{t_e}, \quad U^* = \frac{u}{P_0 h / k^*}, \quad \Sigma_{ii}^* = \frac{\sigma_{ii}}{P_0} \quad (i = r, \theta, z) \quad (42)$$

where $k=10$ GPa and t_e is the loading time duration which was assumed to be 1(s) in Ref. [9].

The numerical results for clamped FG cylindrical shell under the above mentioned conditions are compared with the numerical solution of Setoodeh et al. [9] in Table 3 and Fig. 6. It is obvious that close agreement exists between the results of three approaches. However, the agreement between the present approach and ANSYS is greater than those of LWDQ of Ref. [9] and ANSYS. In addition, it is interesting to note that both ANSYS and LWDQ cannot predict the actual value of the radial stress on the inner surface of the shell, which should be equal to the applied pressure on this surface. However, the present approach gives its exact value.

After demonstrating the convergence and accuracy of the approach, some parametric studies for FG hollow cylinders subjected to pressure and heat flux with moving front edge are performed.

As a first example, the influence of the material graded index (p) on the results for FG cylindrical shells with clamped–clamped and clamped–free ends are shown in Figs. 7 and 8, respectively. It can be seen that the material graded index has significant effect on the radial displacement, tangential and axial normal stress components. However, its effect on the radial component of the stress is small and can be ignored, especially on the point far from the inner surface. In addition, the effect of this parameter on the time histories of the radial displacement and tangential stress components at the point ($\xi=0, \eta=0.5$) are presented in Fig. 9 for FG shell with the both boundary conditions mentioned above. Again one can see that the material graded index (p) considerably affects the response of the FG shell.

The effects of geometrical parameters (thickness and length) on the results of FG cylinder with clamped ends are presented in Figs. 10 and 11, respectively. From Fig. 10, it can be seen that increasing the thickness of the cylinder considerably change the results. Also, it is evident from Fig. 11 that the length of the FG shell has significant effect on the results, except on the radial stress component.

The influence of the length and thickness of the FG shell on the time history of the radial displacement and tangential stress components at point ($\xi=0, \eta=0.5$) of the FG shell are shown in

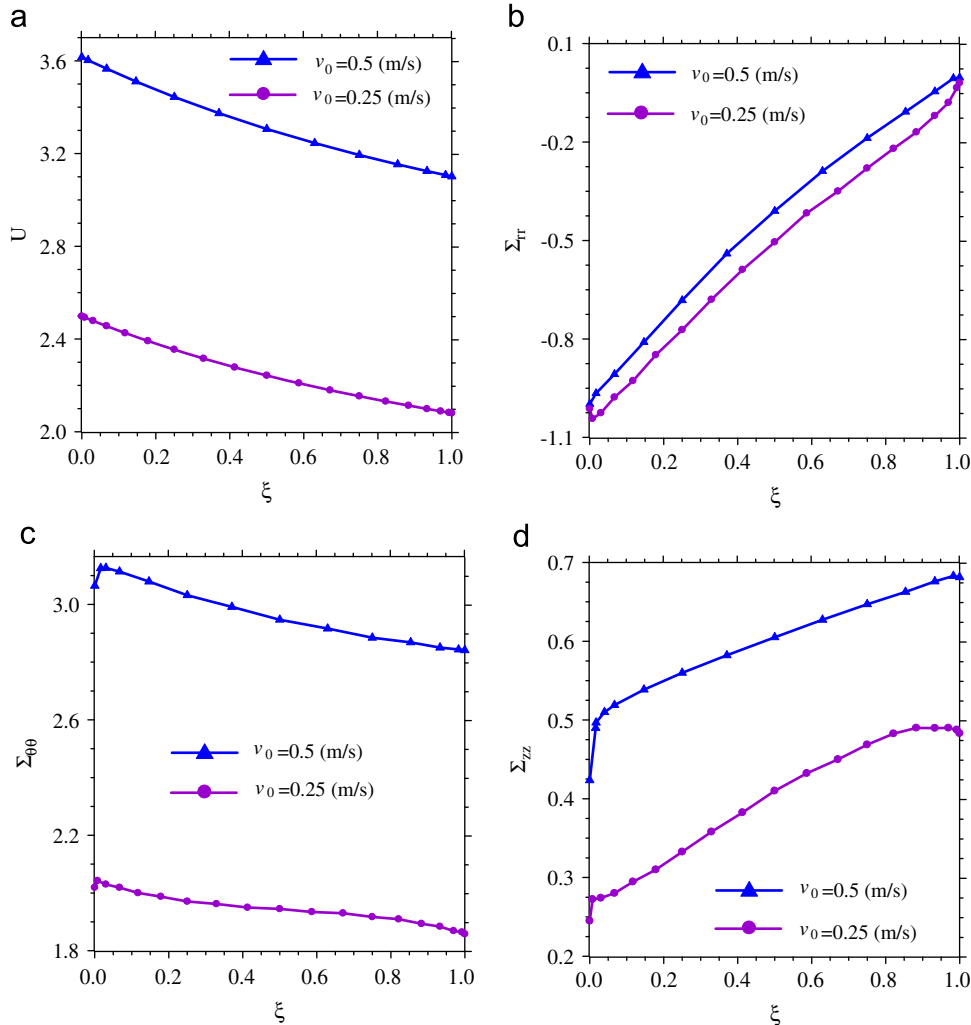


Fig. 13. Comparison of the results for the clamped FG cylindrical shells with two different velocities [$p=1$, $V_e=0.05$, $t=2$ (s), $\eta=0.5$].

Fig. 12. It is noticeable that the length influences the transient part of the solutions much greater than those of steady state portions. But, the thickness has an opposite effect on the response, i.e., its effect increases with increasing elapsed time.

The effect of thermo-mechanical load front edge velocity on the results of the clamped FG cylinder at a given time level is exhibited in Fig. 13. It can be seen that increasing the velocity of the thermo-mechanical load front edge, the non-dimensional parameters at a given point of the cylinder increases. Also, in Fig. 14, the influence of this parameter on the time history of the results at the point ($\xi=0, \eta=0.5$) of the FG shells is presented. It is observable that this parameter has considerable effect on the radial displacement and tangential stress component of the FG cylinder. In addition, with the increase of elapsed time, the difference between the results reduces.

The through the thickness variation of results for the clamped FG cylinder at different time level are compared in Fig. 15. One can see that the parameters approach to their steady state values when the time level increases.

5. Conclusion

The thermoelastic analysis of FG hollow cylinders subjected to moving boundary thermo-mechanical loading is presented. The

formulation is based on the hyperbolic heat conduction equation and elasticity theory for evaluation of temperature and deformation field variables, respectively. The material properties are assumed to be temperature-dependent and graded in the radial (thickness) direction. The governing differential equations are discretized in the longitudinal direction of the FG cylinder using the FEM. Then the resulting system of differential equations and the related boundary conditions are discretized in the radial direction and in their strong form by using DQM. The DQ formulations precisely satisfy the boundary conditions on the inner and outer surfaces of the hollow cylinder. The resulting system of ordinary differential equations is solved in temporal domain using Newmark time integration scheme. The convergence behavior of the method is numerically demonstrated and comparison studies with the available solutions in the literature are performed to validate the presented formulation and the method of solution. Parametric studies are performed to study the transient thermoelastic behavior of FG hollow cylinders under moving thermo-mechanical loading. From the obtained results, it is concluded that the material graded index, the geometrical parameters (length and thickness) and the velocity of the thermo-mechanical loads have significant effects on the radial displacement, tangential and axial stress components of FG hollow cylinders. The influence of thickness on the radial stress component is much greater than the other mentioned parameters.

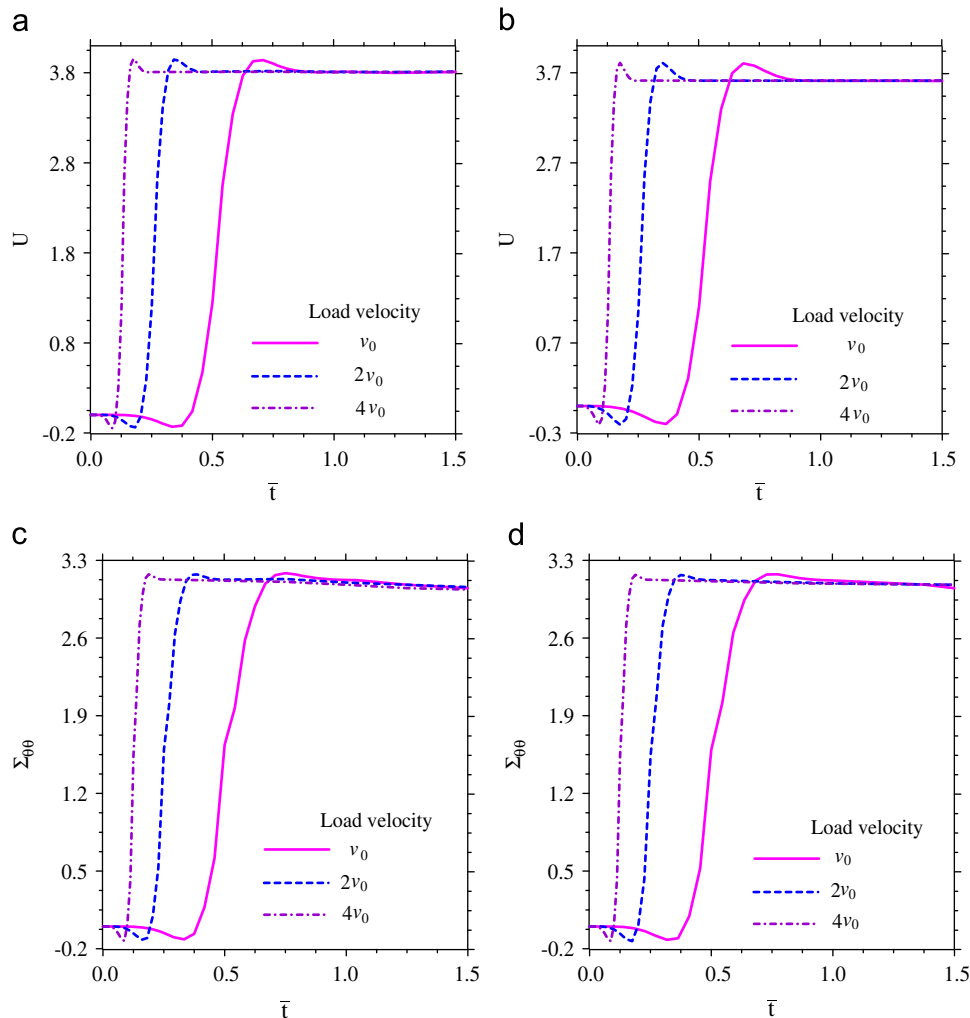


Fig. 14. Influence of the velocity of on the time history of the results at the point ($\xi=0, \eta=0.5$) for the clamped FG cylindrical shells [$V_e=0.05$, $p=1$, $v_0=0.25$ (m/s), $L=1$ (m), $\bar{t}=(v_0 t)/L$].

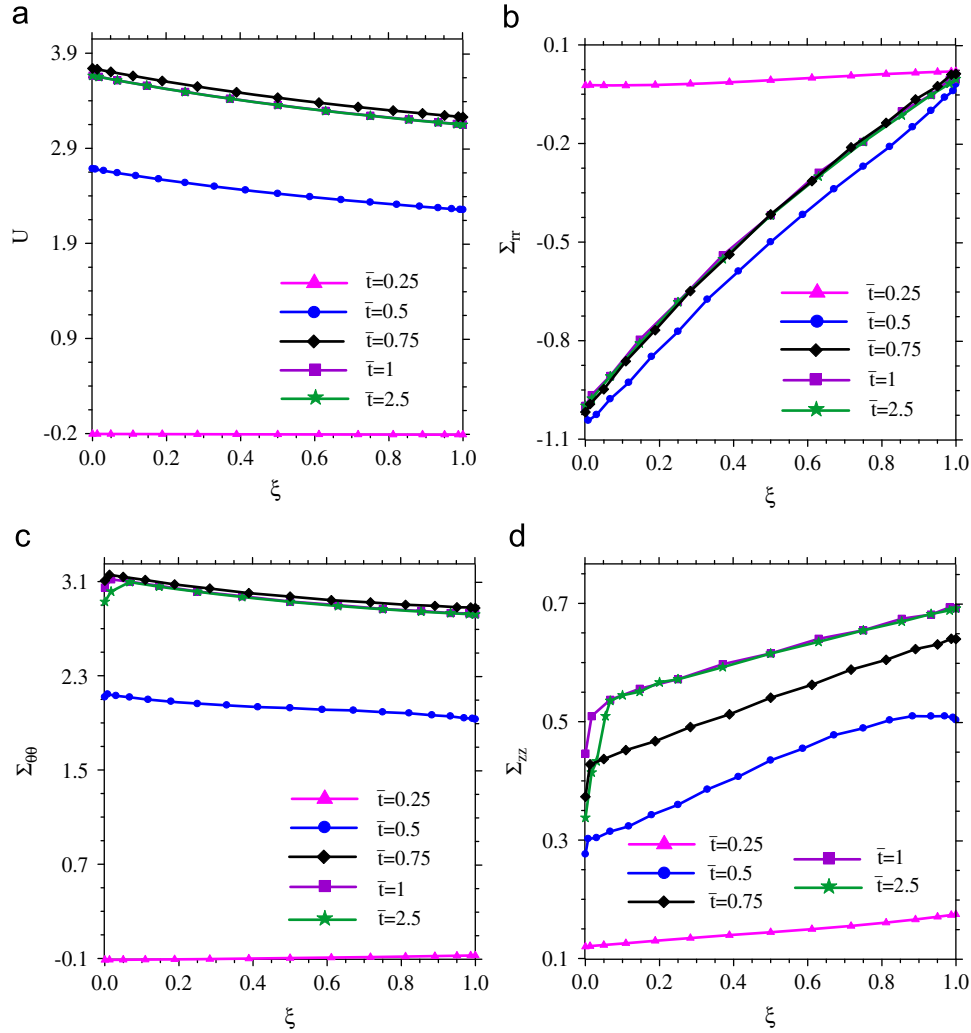


Fig. 15. Comparison of the results for the clamped FG cylindrical shells at different time levels ($p=1$, $V_e=0.05$, $\eta=0.5$).

Appendix. A brief review of DQM

The basic idea of the differential quadrature method is that the derivative of a function with respect to a space variable at a given sampling point, is approximated as the linear weighted sums of its values at all of the sampling points in the domain of that variable. In order to illustrate the DQ approximation, consider a function $f(r, t)$ having its field on a domain $R_i \leq r \leq R_o$ and $0 \leq t$. Let, in the given domain, the function values be known or desired on a grid of sampling point. According to DQ method, the first and second-order derivatives of the function $f(r, t)$ can be approximated as

$$\begin{aligned} \left. \frac{\partial f(r, t)}{\partial r} \right|_{r=r_i} &= \sum_{m=1}^{N_r} A_{im}^r f(r_m, t) = \sum_{m=1}^{N_r} A_{im}^r f_m; \\ \left. \frac{\partial^2 f(r, t)}{\partial r^2} \right|_{r=r_i} &= \sum_{m=1}^{N_r} B_{im}^r f_m \quad \text{for } i = 1, 2, \dots, N_r \end{aligned} \quad (\text{A.1})$$

where A_{ij}^r and B_{ij}^r represent the weighting coefficients of the first and second-order derivatives along the r -direction, respectively. From this equation one can deduce that the important components of DQ approximations are weighting coefficients and the choice of sampling points. In order to determine the weighting coefficients a set of test functions should be used in Eq. (A.1). For polynomial basis functions DQ, a set of Lagrange polynomials are employed as the test functions. The weighting coefficients for the

first-order derivatives in r -direction are thus determined as [23]

$$A_{ij}^r = \begin{cases} \frac{1}{(R_o - R_i)(r_i - r_j)M(r_j)} & \text{for } i \neq j \\ -\sum_{\substack{j=1 \\ i \neq j}}^{N_r} A_{ij}^r & \text{for } i = j; \quad i, j = 1, 2, \dots, N_r \end{cases} \quad (\text{A.2})$$

where $M(r_i) = \prod_{k=1, k \neq i}^{N_r} (r_i - r_k)$.

The weighting coefficients of second order derivative can be obtained as [23]

$$[B_{ij}^r] = [A_{ij}^r][A_{ij}^r] = [A_{ij}^r]^2 \quad (\text{A.3})$$

In numerical computations, Chebyshev–Gauss–Lobatto quadrature points are used, that is [23]

$$r_k = R_i + \frac{(R_o - R_i)}{2} \left\{ 1 - \cos \left[\frac{(k-1)\pi}{(N_r-1)} \right] \right\} \quad \text{for } k = 1, 2, \dots, N_r \quad (\text{A.4})$$

References

- [1] Zimmerman RW, Lutz MP. Thermal stresses and thermal expansion in a uniformly heated functionally graded cylinder. *Journal of Thermal Stresses* 1999;22:177–88.
- [2] Pradhan SC, Loy CT, Lam KY, Reddy JN. Vibration characteristics of functionally graded cylindrical shells under various boundary conditions. *Applied Acoustics* 2000;61:111–29.

- [3] Kim KS, Noda N. Green's function approach to unsteady thermal stresses in an infinite hollow cylinder of functionally graded material. *Acta Mechanica* 2002;156:145–61.
- [4] Kadoli R, Ganesan N. Buckling and free vibration analysis of functionally graded cylindrical shells subjected to a temperature-specified boundary condition. *Journal of Sound and Vibration* 2006;289:450–80.
- [5] Santos H, Mota Soares CM, Mota Soares CA, Reddy JN. A semi-analytical finite element model for the analysis of cylindrical shells made of functionally graded materials under thermal shock. *Composite Structures* 2008;86:10–21.
- [6] Shao ZS, Ma GW. Thermo-mechanical stresses in functionally graded circular hollow cylinder with linearly increasing boundary temperature. *Composite Structures* 2008;83:259–65.
- [7] Tornabene F. Free vibration analysis of functionally graded conical, cylindrical shell and annular plate structures with a four-parameter power-law distribution. *Computer Methods in Applied Mechanics and Engineering* 2009;198:2911–35.
- [8] Peng XL, Li XF. Thermoelastic analysis of a cylindrical vessel of functionally graded materials. *International Journal of Pressure Vessels and Piping* 2010;87:203–10.
- [9] Setoodeh AR, Tahani M, Selahi E. Hybrid layerwise-differential quadrature transient dynamic analysis of functionally graded axisymmetric cylindrical shells subjected to dynamic pressure. *Composite Structures* 2011;93:2663–70.
- [10] Heydarpour Y, Malekzadeh P, Golbahar Haghighi MR, Vaghefi M. Thermo-elastic analysis of rotating laminated functionally graded cylindrical shells using layerwise differential quadrature method. *Acta Mechanica* 2011, <http://dx.doi.org/10.1007/s00707-011-0551-6>.
- [11] Ootao Y, Akai T, Tanigawa Y. Three-dimensional transient thermal stress analysis of a nonhomogeneous hollow circular cylinder due to a moving heat source in the axial direction. *Journal of Thermal Stresses* 1995;18:497–512.
- [12] Hasheminejad SM, Komeili M. Dynamic response of a thick functionally graded material tube under a moving load. *Proceedings of the Institution of Mechanical Engineers, Part C: Journal of Mechanical Engineering and Science* 2007;221:1545–54.
- [13] Jabbari M, Mohazzab AH, Bahtui A. One-dimensional moving heat source in a hollow FGM cylinder. *Journal of Pressure Vessel Technology* 2009;131:021202–8.
- [14] Sheng GG, Wang X. Studies on dynamic behavior of functionally graded cylindrical shells with PZT layers under moving loads. *Journal of Sound and Vibration* 2009;323:772–89.
- [15] Sheng GG, Wang X. Response and control of functionally graded laminated piezoelectric shells under thermal shock and moving loadings. *Composite Structures* 2010;93:132–41.
- [16] Sofiyev AH. Dynamic response of an FGM cylindrical shell under moving loads. *Composite Structures* 2010;93:58–66.
- [17] Wu JJ, Whittaker AR, Cartmell MP. The use of finite element techniques for calculating the dynamic response of structures to moving loads. *Computers and Structures* 2000;78:789–99.
- [18] Ghafoori E, Kargarnovin MH, Ghahremani AR. Dynamic responses of a rectangular plate under motion of an oscillator using a semi-analytical method. *Journal of Vibration and Control* 2010;17:1310–24.
- [19] Ghafoori E, Asghari M. Dynamic analysis of laminated composite plates traversed by a moving mass based on a first-order theory. *Composite Structures* 2010;92:1865–76.
- [20] Malekzadeh P, Farid M, Zahedinejad P, Karami G. Three-dimensional free vibration analysis of thick cylindrical shells resting on two-parameter elastic supports. *Journal of Sound and Vibration* 2008;313:655–75.
- [21] Malekzadeh P. Three-dimensional free vibration analysis of thick functionally graded plates on elastic foundations. *Composite Structures* 2009;89:367–73.
- [22] Malekzadeh P, Alibeygi Beni A. Free vibration of functionally graded arbitrary straight-sided quadrilateral plates in thermal environment. *Composite Structures* 2010;92:2758–67.
- [23] Malekzadeh P. Three-dimensional thermal buckling analysis of functionally graded arbitrary straight-sided quadrilateral plates using differential quadrature method. *Composite Structures* 2011;93:1246–54.
- [24] Reddy JN. An introduction to the finite element method. 3rd ed. New York: McGraw-Hill Higher Education; 2006.